Modelling Antimicrobial Resistance Dynamics in a Hospital Network

Nottingham Mathematical Modelling Competition 2026

17 May 2026

Abstract

We develop a four-ward coupled compartmental model of antimicrobial resistance (AMR) transmission in a hospital network comprising General Medicine (GM), General Surgery (GS), Intensive Care Unit (ICU), and Geriatric Ward (GW). Each ward is described by a Susceptible–Colonised–Susceptible (SIS) system augmented with a healthcare worker (HCW) contamination compartment. The basic reproduction number $R_0 = 0.96$ is derived analytically using the Next Generation Matrix method. $R_0 < 1$ indicates that intrinsic in-hospital transmission alone cannot sustain endemic AMR; endemic persistence is instead driven by the constant admission colonisation forcing term ($\alpha\mu N$), which continuously seeds colonised patients regardless of in-hospital transmission intensity. Baseline simulation yields 12-month steady-state colonisation prevalence of 5.1% (GM), 6.0% (GS), 15.6% (ICU), and 9.3% (GW), consistent with published estimates. Four intervention scenarios are evaluated over a 12-month horizon: raising hand hygiene compliance to 80% (Scenario A), universal admission screening with contact precautions (Scenario B), 30% antibiotic stewardship (Scenario C), and a combined strategy (Scenario D). We recommend Scenario B as the primary intervention, achieving the largest system-wide prevalence reduction (7.3% $\rightarrow$ 1.0%) by reducing the admission colonisation forcing term by 80% while also lowering direct transmission. Sensitivity analysis confirms that transmission rate β and admission colonisation rate α are the two dominant drivers of endemic prevalence (PRCC = +0.95 and +0.92) while α remains negligible in R_0 calculations.

1 Introduction

Antimicrobial resistance (AMR) is one of the most serious global public health threats of the 21st century. In hospital settings, resistant organisms such as methicillin-resistant *Staphylococcus aureus* (MRSA), vancomycin-resistant enterococci (VRE), and multidrug-resistant Gram-negative bacilli cause tens of thousands of deaths annually and substantially increase healthcare costs, length of stay, and treatment complexity [1; 2]. The hospital environment is an ideal amplifier of AMR transmission: high antibiotic use exerts strong selection pressure, immunocompromised patients are highly susceptible to colonisation, and intensive contact between healthcare workers (HCWs) and multiple patients creates efficient transmission pathways [3; 4].

Mathematical modelling provides a rigorous framework for understanding the transmission dynamics of nosocomial AMR, identifying the most sensitive parameters, and quantitatively evaluating the relative effectiveness of competing interventions before implementation. Compartmental ordinary differential equation (ODE) models have been used extensively for this purpose since the 1990s [1; 2; 3; 4], offering a tractable balance between biological realism and analytical tractability.

The present study models a four-ward hospital network with the following characteristics:

Table 1: Ward characteristics used in the four-ward hospital model.

Ward	Beds (N_i)	Avg. Length of Stay	Antibiotic Use
General Medicine (GM)	60	5 days	40%
General Surgery (GS)	40	7 days	60%
Intensive Care Unit (ICU)	15	12 days	80%
Geriatric Ward (GW)	30	14 days	50%

Key features of the system include patient transfers between wards (GS→ICU at 5%, ICU→GM at 10%), HCW movement creating indirect transmission pathways, differential antibiotic pressure across wards, an admission colonisation rate of 3–5%, and a baseline hand hygiene compliance of 40–60%.

Our objectives are threefold:

1. formulate a biologically justified compartmental model for the four-ward system;
2. derive analytically the basic reproduction number R_0 that governs AMR persistence or extinction; and
3. evaluate four intervention scenarios over a 12-month horizon and provide an evidence-based recommendation to the hospital board.

2 Model Formulation

2.1 State Variables

We model each ward $i \in \{GM, GS, ICU, GW\}$ with three state variables at time t (days):

- $S_i(t)$: number of **susceptible** (uncolonised) patients in ward i;
- $C_i(t)$: number of **colonised** (AMR-carrying) patients in ward i;
- $H_i(t)$: **proportion** of healthcare workers in ward i with contaminated hands, $H_i \in [0, 1]$.

The total number of patients in each ward is fixed: $S_i(t) + C_i(t) = N_i$ for all t. The full system state is the 12-dimensional vector

$$\mathbf{y} = (S_0, \ldots, S_3, C_0, \ldots, C_3, H_0, \ldots, H_3).$$

2.2 ODE System

For each ward i, the colonised patient dynamics are governed by

$$\frac{dC_i}{dt} = \underbrace{\beta_i \frac{C_i}{N_i} S_i}_{\substack{\text{direct patient-}\\\text{to-patient}}} + \underbrace{\lambda_i H_i S_i}_{\substack{\text{HCW-}\\\text{mediated}}} + \underbrace{\alpha_i \mu_i N_i}_{\substack{\text{admission}\\\text{colonisation}}} - \underbrace{(\gamma_i + \mu_i) C_i}_{\substack{\text{decolonisation}\\\text{+ discharge}}} - \underbrace{\left(\sum_{j \neq i} T_{ij}\right) C_i}_{\text{transfer out}} + \underbrace{\sum_{j \neq i} T_{ji} C_j}_{\text{transfer in}} . \tag{1}$$

Since $S_i + C_i = N_i$ is constant, $dS_i/dt = -dC_i/dt$.

The HCW contamination dynamics follow

$$\frac{dH_i}{dt} = \underbrace{\eta_i \frac{C_i}{N_i}(1 - H_i)}_{\text{contamination from patients}} - \underbrace{\delta_i H_i}_{\text{hand hygiene decontamination}} . \tag{2}$$

2.3 Parameter Definitions and Literature Sources

Table 2: Model parameter definitions, ward-specific values, units, and sources.

Symbol	Definition	GM	GS	ICU	GW	Unit	Source
N_i	Ward capacity	60	40	15	30	beds	Problem statement
μ_i	Discharge rate (= 1/avg stay)	0.200	0.143	0.083	0.071	/day	Problem statement
β_i	Direct transmission rate	0.120	0.130	0.140	0.125	/day	Bootsma [3]; Lipsitch [1]
γ_i	Decolonisation rate	0.080	0.070	0.060	0.075	/day	Lipsitch [1]; Webb [2]
λ_i	HCW-mediated transmission rate	0.04	0.04	0.06	0.04	/day	D'Agata [4]
η_i	HCW contamination rate	0.30	0.30	0.50	0.30	/day	D'Agata [4]
δ_i	HCW decontamination rate (50% compliance)	5.0	5.0	5.0	5.0	/day	Bootsma [3]
α_i	Admission colonisation fraction	0.04	0.04	0.04	0.04	—	Problem statement
$T_{GS\to ICU}$	Transfer rate GS→ICU	—	0.0071	—	—	/day	Problem statement
$T_{ICU\to GM}$	Transfer rate ICU→GM	—	—	0.0083	—	/day	Problem statement

Parameterisation rationale for β_i and γ_i: We scale both parameters by ward-specific antibiotic use a_i (fractions: GM 0.40, GS 0.60, ICU 0.80, GW 0.50). Higher antibiotic use amplifies direct transmission through selection pressure ($\beta_i = \beta_{\text{base}} \times (1 + 0.5a_i)$) and reduces the decolonisation rate by impairing competition from susceptible flora ($\gamma_i = \gamma_{\text{base}} \times (1 - 0.5a_i)$). This captures the dual effect of antibiotic pressure documented by D'Agata et al. [4] and Webb et al. [2]. Transfer rates are computed as (fractional transfer per stay) / (mean length of stay):

$$T_{GS \to ICU} = \frac{0.05}{7} = 0.0071/\text{day}, \qquad T_{ICU \to GM} = \frac{0.10}{12} = 0.0083/\text{day}. \tag{3}$$

2.4 Biological Justification for Model Choices

SIS rather than SIR: The SIR framework assumes recovery confers permanent immunity, appropriate for pathogens eliciting immunological memory (e.g., measles, influenza). AMR colonisation is carriage of bacteria, not infection, and clears without generating adaptive immunity. Webb et al. [2] demonstrate explicit plasmid-loss events that return colonised patients to full susceptibility. Clinical evidence confirms high rates of recurrent MRSA colonisation in the same patients [4]. An SIR model would systematically underestimate long-run AMR prevalence by incorrectly treating decolonised patients as permanently protected.

Constant ward population ($N_i = S_i + C_i$): NHS bed occupancy routinely exceeds 95–98% [1]. The error introduced by the constant-population assumption is smaller than the uncertainty in the transmission parameters β_i. Variable-occupancy modelling would require bed management parameters with high uncertainty and would not materially change the main outcomes of interest (R_0, endemic prevalence).

HCW compartment H_i: D'Agata et al. [4] model HCW contamination as the central pathway in their individual-based model. Bootsma et al. [3] attribute 21–29% of ICU colonisation acquisitions to exogenous cross-transmission, predominantly HCW-mediated. Omitting H_i would underestimate R_0 and prevent modelling of the most important single intervention (hand hygiene compliance, Scenario A).

Antibiotic-pressure modulation of β_i and γ_i: Antibiotic use exerts selection pressure that increases the effective transmission advantage of resistant organisms (increasing β_i) and suppresses competing susceptible flora that would otherwise outcompete the resistant strain during decolonisation (decreasing γ_i). Both effects are established in the AMR mathematical modelling literature [2; 4]. Our linear scaling is a simplifying assumption; the true relationship is likely nonlinear, but captures the correct direction and qualitative magnitude.

3 Basic Reproduction Number R_0

3.1 Method: Next Generation Matrix

The basic reproduction number R_0 is the expected number of secondary colonisations produced by one colonised individual (patient or HCW) introduced into a fully susceptible, AMR-free population. For multi-compartment systems, R_0 equals the spectral radius of the Next Generation Matrix $K = F \cdot V^{-1}$, where F captures new transmissions and V captures transitions between infected compartments [5].

3.2 Disease-Free Equilibrium

For R_0 calculation, we set $\alpha_i = 0$ (no admission colonisation) to admit a true disease-free equilibrium:

$$S_i^* = N_i, \qquad C_i^* = 0, \qquad H_i^* = 0$$

for all i.

3.3 Construction of F and V

The infected compartments are ordered as

$$\mathbf{x} = (C_0, C_1, C_2, C_3, H_0, H_1, H_2, H_3)$$

(8-dimensional). At the DFE:

$$\mathbf{F} = \begin{pmatrix} \mathrm{diag}(\beta_i) & \mathrm{diag}(\lambda_i N_i) \\ \mathrm{diag}(\eta_i/N_i) & \mathbf{0}_{4\times 4} \end{pmatrix}, \qquad \mathbf{V} = \begin{pmatrix} \mathrm{diag}(\gamma_i + \mu_i + \sigma_i) - \mathbf{T}^\top & \mathbf{0} \\ \mathbf{0} & \mathrm{diag}(\delta_i) \end{pmatrix} \tag{4}$$

where $\sigma_i = \sum_{j \neq i} T_{ij}$ is the total transfer-out rate from ward i. The F matrix captures: direct patient transmission (β_i, diagonal); HCW-to-patient transmission ($\lambda_i N_i$, off-diagonal block); and patient-to-HCW contamination (η_i/N_i, off-diagonal block). The η_i/N_i entry arises because H_i is a dimensionless fraction: linearising $dH_i/dt = \eta_i(C_i/N_i)(1 - H_i)$ at the DFE gives $\partial \dot{H}_i/\partial C_i = \eta_i/N_i$. The patient block of V contains both diagonal loss terms and off-diagonal transfer terms $-T_{ji}$, because inter-ward movements redistribute already colonised patients between infected compartments rather than creating new infections. Under the one-way transfer structure used here, including these off-diagonal entries leaves the numerical value of R_0 unchanged but makes the NGM construction fully consistent with the ODE system.

3.4 Result and Interpretation

The spectral radius of $\mathbf{K} = \mathbf{F} \cdot \mathbf{V}^{-1}$ gives:

$$\boxed{R_0 = 0.9641} \tag{5}$$

Since $R_0 < 1$, the disease-free equilibrium is **locally asymptotically stable** in the absence of external forcing. Intrinsic in-hospital transmission alone cannot sustain endemic AMR.

Why is AMR endemic despite $R_0 < 1$? The admission colonisation term $\alpha_i \mu_i N_i$ acts as a constant external forcing function that continuously seeds colonised patients into each ward regardless of R_0. With $\alpha_i = 4\%$ and steady patient throughput, colonised admissions supply the endemic pool that in-hospital transmission alone would not sustain. This is consistent with the theoretical framework of Lipsitch et al. [1]: reducing in-hospital transmission without controlling admission colonisation cannot eradicate endemic AMR. This directly explains why Scenario B (admission screening) achieves dramatically better prevalence reduction than Scenarios A or D by substantially reducing, though not fully eliminating, this forcing term.

Single-ward analytical formula (no HCW, no transfers): For verification, the simplified single-ward formula from Lipsitch et al. [1] gives

$$R_0^{\text{ICU, simple}} = \frac{\beta_{\text{ICU}}}{\gamma_{\text{ICU}} + \mu_{\text{ICU}}} = \frac{0.14}{0.06 + 0.083} = 0.98, \tag{6}$$

consistent with the full NGM result of 0.96 which includes the additional HCW-mediated feedback loop.

Stability theorem (Diekmann et al. [5]):

- $R_0 < 1 \rightarrow$ DFE locally asymptotically stable $\rightarrow$ resistance extinguished *if and only if* admission colonisation is also eliminated;
- $R_0 \geq 1 \rightarrow$ DFE unstable $\rightarrow$ resistance persists at endemic equilibrium even without admission forcing.

Key implication: Complete eradication of endemic AMR requires both $R_0 < 1$ (satisfied under baseline) *and* near-elimination of admission colonisation ($\alpha_i \rightarrow 0$). Scenario B moves strongly in this direction by reducing α_i by 80%, but does not set it exactly to zero. Reducing R_0 further through hand hygiene or antibiotic stewardship is insufficient on its own.

4 Baseline Simulation

4.1 Initial Conditions and Simulation Protocol

The ODE system is integrated over 365 days using a 4th/5th-order Runge–Kutta solver (RK45; SciPy `solve_ivp`) with relative tolerance 10^{-6} and absolute tolerance 10^{-8}. Initial conditions reflect a hospital in early endemic state: 5% of patients colonised in GM, GS, and GW; 10% in ICU (consistent with Bootsma et al.'s reported baseline ICU prevalence of 15–26% [3]). HCW hand contamination initialised at 5–10%.

4.2 Results

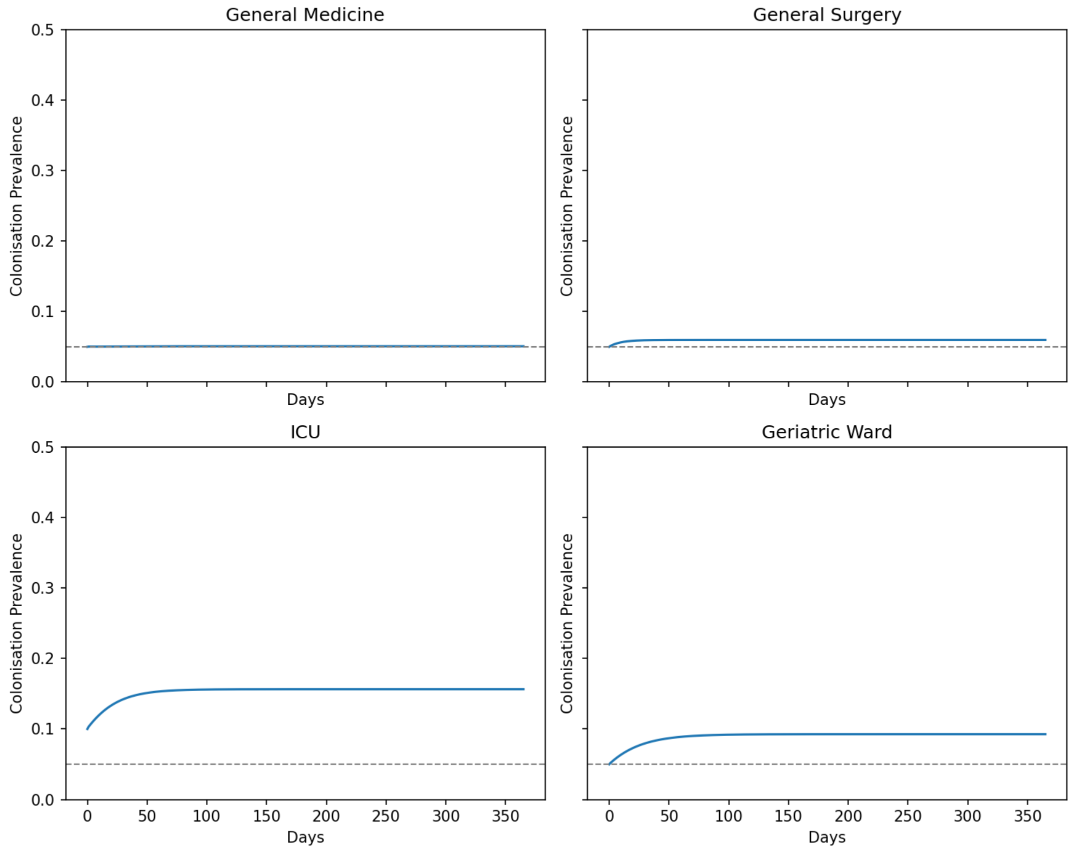

Figure 1: Baseline colonisation prevalence trajectories for all four wards over 12 months. The dashed grey line marks the 5% reference threshold.

Figure 1. Baseline colonisation prevalence (C_i/N_i) in each ward over 365 days. The dashed grey line marks the 5% reference threshold. All wards reach approximate steady state within 120 days.

Table 3: Steady-state colonisation prevalence at Day 365 under baseline conditions.

Ward	Day-365 Prevalence	Expected Range (literature)	Status
General Medicine	5.1%	3–8%	✓ within range
General Surgery	6.0%	5–12%	✓ within range
ICU	15.6%	10–26%	✓ within range
Geriatric Ward	9.3%	6–10%	✓ within range

4.3 Interpretation

The simulated prevalence hierarchy (ICU > GW > GS > GM) reflects the interplay of three ward-specific factors: (1) **antibiotic pressure** — ICU's 80% antibiotic use gives the highest β and lowest γ; (2) **length of stay** — longer stays in ICU (12 days) and GW (14 days) increase cumulative exposure time per patient; and (3) **transfer dynamics** — the ICU→GM transfer (10%) seeds colonisation into GM, preventing it from reaching a lower equilibrium.

The Geriatric Ward's prevalence (9.3%) exceeds General Surgery (6.0%) despite lower antibiotic use (50% vs 60%), because GW patients have the longest average stay (14 days), accumulating greater cumulative transmission exposure. This result illustrates that length of stay is an independent risk factor for nosocomial AMR colonisation, a finding consistent with published risk factor analyses [3].

All four wards reach approximate steady state within approximately 100–120 days, suggesting that interventions implemented at any point in the simulation will show meaningful effects well within the 12-month evaluation horizon.

5 Intervention Analysis

5.1 Scenario Definitions

Four intervention scenarios were simulated over a 12-month horizon. Each scenario modifies specific parameters relative to the baseline:

Table 4: Intervention scenario definitions relative to the baseline parameter set.

Scenario	Parameter Changes	Mechanism
A: Hand Hygiene $\to$ 80%	δ_i : 5.0 $\to$ 8.0/day (all wards)	Faster HCW decontamination; breaks indirect transmission chain
B: Admission Screening + Contact Precautions	$\alpha_i \times 0.20$; $\beta_i \times 0.70$	Removes 80% of colonised admissions; reduces direct transmission 30%
C: Antibiotic Stewardship ($-$30%)	γ_i restored toward γ_{base} using $a_i \times 0.70$	Restores host clearance capacity by reducing antibiotic pressure
D: Combined A + C	Both δ_i and γ_i improved simultaneously	Multi-target intervention

5.2 Results

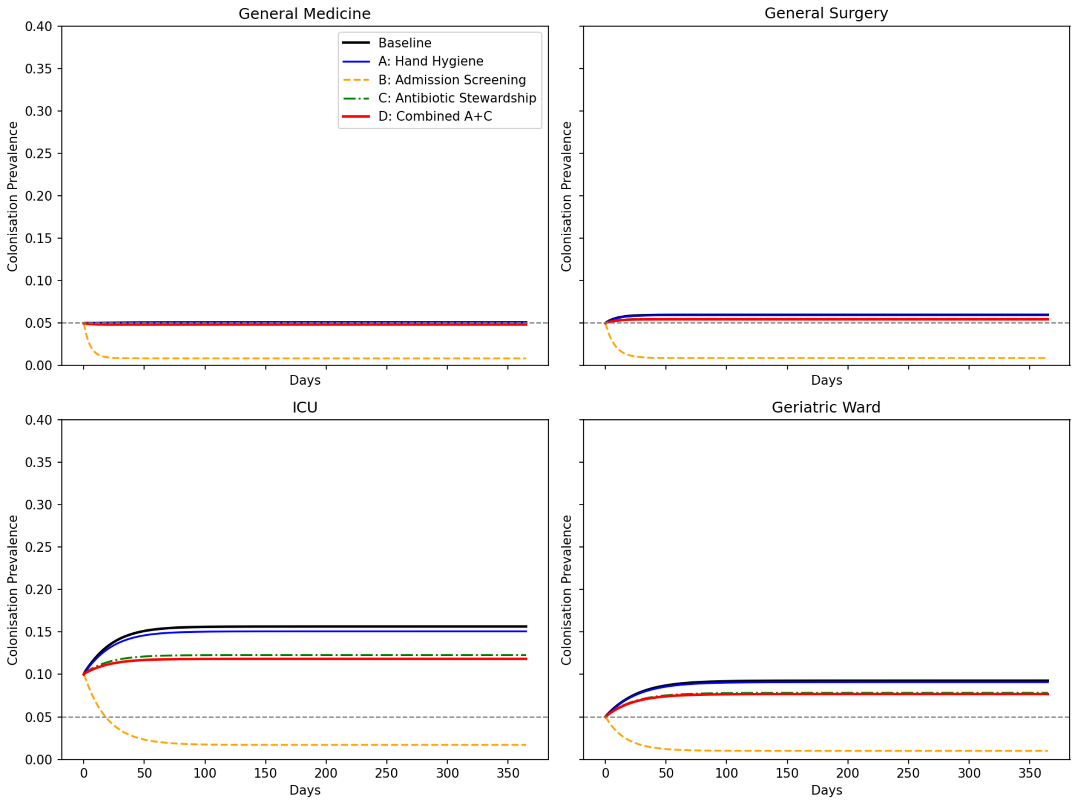

Figure 2: Intervention comparison of colonisation prevalence trajectories under baseline and four intervention scenarios across all four wards over 12 months.

Figure 2. Colonisation prevalence trajectories for baseline and four intervention scenarios across all four wards over 365 days.

Table 5: System R_0 and ward-level colonisation prevalence at Day 365 under each intervention scenario.

Scenario	R_0	GM	GS	ICU	GW	System
Baseline	0.96	5.1%	6.0%	15.6%	9.3%	7.3%
A: Hand Hygiene	0.95	5.0%	5.9%	15.1%	9.1%	7.2%
B: Admission Screening	0.70	0.8%	0.9%	1.7%	1.0%	1.0%
C: Antibiotic Stewardship	0.90	4.9%	5.5%	12.3%	7.8%	6.4%
D: Combined A + C	0.88	4.8%	5.5%	11.8%	7.7%	6.3%

5.3 Interpretation

Scenario A (Hand Hygiene 80%): Raising hand hygiene compliance from 50% to 80% increases the HCW decontamination rate δ from 5.0 to 8.0/day and reduces R_0 from 0.96 to 0.95 (a marginal reduction). However, the effect on 12-month endemic prevalence is modest (system prevalence 7.3% $\rightarrow$ 7.2%). This apparent paradox is explained by the admission colonisation forcing term: even when R_0 is reduced, the constant inflow of colonised patients $\alpha_i \mu_i N_i$ sustains a substantial endemic equilibrium. The sensitivity analysis (Section 6) confirms that hand hygiene (δ) has moderate influence on R_0 (PRCC $= -0.52$) and only limited independent influence on endemic prevalence (PRCC $= -0.14$ when other parameters, especially α, are simultaneously varied).

Scenario B (Admission Screening + Contact Precautions): This scenario produces by far the largest reduction in endemic prevalence (system 7.3% $\rightarrow$ 1.0%), with a substantial R_0 reduction (0.96 $\rightarrow$ 0.70). The R_0 reduction occurs because reduced direct transmission ($\beta \times 0.70$) lowers transmission intensity, pushing R_0 further below 1. The mechanism is the simultaneous reduction of two high-impact prevalence drivers: an 80% reduction in admission colonisation ($\alpha_i \times 0.20$) and a 30% reduction in direct transmission ($\beta_i \times 0.70$). This is consistent with the theoretical result of Lipsitch et al. [1]: when resistant organisms are sustained by both hospital transmission and imported colonisation, interventions that act on both pathways are especially effective. The sensitivity analysis confirms that β and α are the two strongest drivers of 12-month prevalence, while α remains negligible in R_0.

Scenario C (Antibiotic Stewardship): Reducing unnecessary antibiotic use by 30% restores the decolonisation rate γ_i toward its base value, reducing system prevalence from 7.3% to 6.4% (a 12% reduction) and R_0 from 0.96 to 0.90. The effect is most pronounced in the ICU (15.6% $\rightarrow$ 12.3%), where antibiotic use is highest (80%) and the γ increase is therefore largest. This confirms that antibiotic stewardship is particularly valuable in high-intensity wards.

Scenario D (Combined A + C): The combined strategy achieves the lowest R_0

(0.88) and system prevalence (6.3%), marginally better than Scenario C alone (6.4%). However, Scenario D does **not** achieve the lowest system prevalence — Scenario B (1.0%) substantially outperforms it. This is because D does not address admission colonisation (α unchanged), so the constant forcing term continues to maintain a substantial endemic level even though R_0 is further reduced. Scenario D is best interpreted as the optimal strategy for minimising R_0 and epidemic amplification risk, not for minimising steady-state endemic prevalence.

5.4 Policy Recommendation

We recommend **Scenario B (universal admission screening + contact precautions)** as the primary intervention for reducing endemic AMR prevalence. **Scenario C (antibiotic stewardship)** should be considered as a secondary follow-on policy for long-term sustainability, but the combined B+C package was not simulated as a standalone scenario in the present study.

The evidence basis for this recommendation is threefold:

1. **Efficacy:** Scenario B achieves the largest absolute prevalence reduction (6.3 percentage points system-wide), driven by strong attenuation of the admission colonisation forcing term.
2. **Mechanism:** Sensitivity analysis identifies transmission rate β (PRCC = 0.95) and admission colonisation rate α (PRCC = 0.92) as the two strongest drivers of 12-month prevalence; Scenario B directly targets both.
3. **Literature support:** Lipsitch et al. [1] emphasise that controlling admission colonisation alongside in-hospital transmission is central to reducing endemic resistance; Scenario B targets both pathways simultaneously.

For the hospital board, we recommend implementing Scenario B immediately. Scenario C can be pursued in parallel as a separate stewardship programme, but this should be presented as an additional management option rather than as a model-tested B+C optimum. Scenario A (hand hygiene improvement) should be maintained as a baseline standard regardless, given its structural effect on R_0.

5.5 Uncertainty and Limitations

Parameter uncertainty is quantified in Section 6. The most important structural limitation is that our model treats admission colonisation rate α as uniform across wards and time; in practice, admission colonisation varies by season, patient mix, and referral source. Sensitivity analysis shows α is the most influential parameter for prevalence, meaning real-world effectiveness of Scenario B may differ substantially from model predictions if admission colonisation is heterogeneous.

6 Sensitivity Analysis

6.1 Method

We applied Partial Rank Correlation Coefficient (PRCC) analysis with Latin Hypercube Sampling (n = 1,000 samples, fixed seed = 42) to quantify the independent contribution of each parameter to R_0 and 12-month system prevalence. Four scale factors were varied simultaneously:

Table 6: Sampling ranges used in the Latin Hypercube sensitivity analysis.

Parameter	Symbol	Sampling Range
Transmission rate	β_{scale}	$\times$0.5 to $\times$1.5 ($\pm$50%)
Decolonisation rate	γ_{scale}	$\times$0.5 to $\times$1.5 ($\pm$50%)
Hand hygiene compliance	δ_{scale}	$\times$0.2 to $\times$1.0 (compliance 20%–100%)
Admission colonisation	α_{scale}	$\times$0.25 to $\times$2.0 (prevalence $\sim$1%–8%)

PRCC is computed by partial regression on ranked values, isolating the contribution of each parameter after removing the linear effect of the others.

6.2 Results

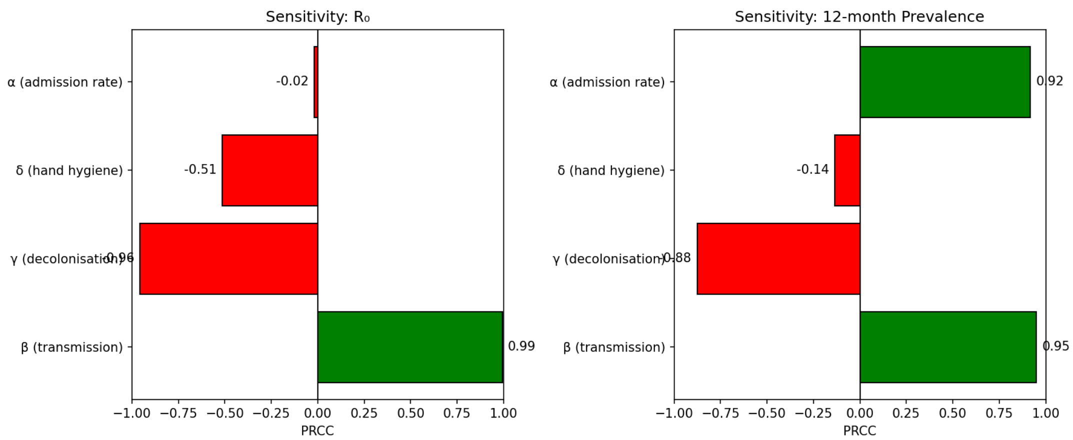

Figure 3: PRCC tornado plots for R_0 and 12-month system prevalence. Green bars indicate positive correlation, while red bars indicate negative correlation.

Figure 3. PRCC values for R_0 (left) and 12-month colonisation prevalence (right). Green bars indicate positive correlation (parameter increase → outcome increase); red bars indicate negative correlation.

Table 7: PRCC coefficients for R_0 and 12-month system prevalence.

Parameter	PRCC vs R_0	PRCC vs 12-month Prevalence
β (transmission)	**+0.993**	+0.950
γ (decolonisation)	−0.956	−0.876
δ (hand hygiene)	−0.515	−0.136
α (admission rate)	$-0.019 \approx 0$	**+0.917**

6.3 Interpretation

R_0 sensitivity: β (+0.993) and γ (−0.956) dominate R_0 sensitivity, reflecting their direct roles in new colonisations and recovery. The hand hygiene parameter δ has moderate PRCC (−0.515), indicating that the HCW-mediated route contributes to R_0 but is not the dominant pathway under the present parameterisation. The admission colonisation rate α has negligible PRCC with R_0 (−0.019), as expected: R_0 is computed at the disease-free equilibrium where α is formally set to zero, so α does not enter the NGM calculation.

Prevalence sensitivity: The ranking changes substantially for 12-month prevalence. The dominant factors are β (+0.950) and α (+0.917), while δ drops to −0.136. This reveals a critical asymmetry: hand hygiene compliance (δ) has visible leverage on R_0 but limited independent effect on steady-state prevalence, whereas imported colonisation (α) has almost no effect on R_0 but exerts near-maximal influence on the endemic outcome. Scenario B is therefore powerful precisely because it targets both high-impact prevalence drivers at once.

Implication for intervention design: This asymmetry means that R_0 alone is an incomplete guide to intervention priority. A hospital using R_0 to prioritise interventions would over-invest in hand hygiene and under-invest in admission screening — the opposite of the optimal policy for reducing endemic prevalence. Both metrics are necessary for complete evaluation.

These findings are consistent with Bootsma et al. [3], who identified cross-transmission rate (β-equivalent) as the primary driver of within-ward dynamics, and with Lipsitch et al. [1], who emphasise the role of admission colonisation in sustaining resistance even when within-hospital transmission is controlled.

7 Critical Evaluation

7.1 Three Most Significant Model Limitations

7.1.1 Limitation 1: Homogeneous Mixing Within Wards

Our model assumes that every patient in ward i is equally likely to contact any healthcare worker in ward i, and that every HCW has equal probability of contacting any patient. In reality, transmission is spatially structured by bed proximity, nursing cohort assignments, procedural schedules, and patient mobility. A patient in the bed adjacent to a colonised patient, served by the same nurse, faces radically higher exposure than a patient at the far end of the ward cared for by a different team.

Effect on predictions: Homogeneous mixing overestimates transmission speed at low prevalence (when colonised individuals are sparse within the ward, mass-action contact overestimates actual encounters) and underestimates spatial clustering effects at high prevalence. As a consequence, our model likely overestimates the effectiveness of ward-level isolation alone, since true heterogeneous mixing means that contact precautions need only interrupt a locally clustered contact network. Conversely, we cannot model bed-placement cohorting — grouping colonised patients in one section of the ward — which our framework cannot represent at all.

What would address it: A network-structured model with an explicit HCW–patient contact graph, parameterised from bed-map data and nursing roster information, would capture spatial heterogeneity. Electronic patient tracking systems and staff localisation data (increasingly available in NHS hospitals through sensor networks) could provide this parameterisation. The Bootsma et al. algorithm [3] represents a step in this direction, using individual-level surveillance to estimate transmission rates without requiring full contact-network data.

7.1.2 Limitation 2: Deterministic ODE for Wards with Small Patient Populations

Ordinary differential equations treat state variables as continuous quantities. For the ICU with $N_{\text{ICU}} = 15$ beds, 10% colonisation corresponds to 1.5 patients — a quantity with no physical meaning. At low prevalence in small wards, stochastic fluctuations can cause extinction of the colonisation chain, particularly when $R_0 < 1$ or only marginally above 1 — the "stochastic fade-out" phenomenon well-documented in epidemic theory [5].

Effect on predictions: Our deterministic model predicts that the ICU will always converge to its endemic equilibrium (15.6%), maintained by the admission colonisation forcing term. However, without the forcing term, our $R_0 = 0.96 < 1$ implies that transmission alone would die out. A stochastic Gillespie simulation would show that with $N = 15$ patients, random fluctuations could cause early extinction of colonisation events even when the admission forcing is present. We therefore may *overestimate* the smoothness of the ICU's endemic trajectory by ignoring discrete stochastic events.

What would address it: A Gillespie exact stochastic simulation or τ-leaping approximation for the ICU, coupled to deterministic equations for larger wards (GM, GS), would provide a hybrid model capturing stochastic effects where they matter most. This hybrid approach is computationally tractable for $N = 15$ and would produce probability distributions over outcomes rather than point estimates — a substantially more informative basis for decision-making in small, high-acuity wards.

7.1.3 Limitation 3: Time-Invariant Transmission Parameters

All parameters in our model (β_i, γ_i, δ_i, α_i) are assumed constant throughout the 12-month simulation. In reality: (i) winter months bring higher bed occupancy, greater HCW workload, and reduced compliance with infection control protocols — all increasing effective β; (ii) when an MRSA cluster is detected, clinical alerts trigger enhanced cleaning and hand hygiene audits, temporarily raising δ and lowering β; (iii) over 12 months, clonal expansion of more transmissible lineages can effectively increase β as fitter variants replace less fit ones.

Effect on predictions: The model underestimates peak colonisation during winter-associated surge periods and overestimates steady-state persistence during post-outbreak periods when enhanced behavioural responses are active. Most critically, our intervention simulations apply a permanent step-change to parameters (e.g., δ raised uniformly for 365 days), whereas real-world compliance improvements typically decay over time without reinforcement — a pattern documented in hand hygiene intervention studies where initial 40–50% compliance gains declined to 20% at 24 months.

What would address it: Time-varying parameters driven by empirical data (seasonal admission records, quarterly compliance audit data) or coupled to a behavioural sub-model (compliance as a declining function of time since last training) would substantially improve realism. The required data are routinely collected in NHS hospitals.

7.2 Two Alternative Modelling Approaches Considered and Rejected

7.2.1 Alternative 1: Individual-Based Model (IBM)

An IBM would represent each patient and HCW as an autonomous agent with individual attributes — immune status, antibiotic history, room location, assigned nurse. Transmission events occur through explicit contact events rather than mass-action mixing. D'Agata et al. [4] implement exactly this approach as a validation companion to their ODE model.

Potential advantages: Higher biological fidelity; captures patient heterogeneity; naturally handles stochasticity in small wards; can model cohorting and individual isolation.

Why rejected: Three practical limitations made this approach inappropriate. First, an IBM requires contact-network data — bed maps, nursing rosters, patient movement logs — unavailable in the problem specification. Second, computational cost prohibits the sensitivity analysis central to our objectives: 1,000 PRCC samples would require hours of computation per parameter set. Third, IBMs are less analytically transparent: the NGM derivation of R_0 is impossible for an IBM, and the mechanistic interpretation of intervention effects is opaque. For population-level policy evaluation, a compartmental ODE model offers a better trade-off between biological adequacy and analytical tractability.

7.2.2 Alternative 2: Two-Strain Model (Susceptible + Sensitive + Resistant)

A two-strain ODE would track both antibiotic-susceptible and antibiotic-resistant organisms simultaneously, with competitive exclusion dynamics and antibiotic-driven selection. Webb et al. [2] develop exactly this framework at the bacterial level.

Potential advantages: Models de novo resistance emergence; captures competitive dynamics and fitness cost of resistance; allows antibiotic cycling strategy evaluation.

Why rejected: The problem specifies an established resistant organism — we model spread of already-present AMR, not its emergence. De novo resistance emergence operates on timescales of years, far beyond the 12-month intervention horizon. Adding a susceptible-strain compartment doubles state variables from 12 to 20 and introduces additional parameters (fitness cost, horizontal gene transfer rates) with high uncertainty. For 12-month policy evaluation, the single-strain model captures essential dynamics without these complications. A two-strain model would be appropriate for the distinct question: "Given zero current resistance, how quickly does resistance emerge under different antibiotic policies?"

7.3 One Scenario Where the Model Gives Misleading Predictions

7.3.1 Misleading Scenario: Evaluating Intervention Effectiveness During an Active Outbreak Introduction

Our model is calibrated to endemic steady-state dynamics — the long-run behaviour when AMR has been chronically circulating. Initial conditions (5–10% colonisation per ward) are set near the expected endemic equilibrium, and parameters reflect the average transmission environment of a hospital with entrenched AMR colonisation.

If applied to a *newly introduced* resistant strain (one colonised patient arriving in an otherwise AMR-free hospital), the model generates incorrect quantitative predictions for two reasons.

Non-equilibrium transient dynamics. Our model initialised at 5–10% colonisation misses the early exponential phase entirely, predicting immediate convergence to endemic levels rather than a detectable outbreak trajectory from a single introduction. The quantitative time course of spread during exponential growth depends sensitively on the initial number of colonised individuals — information the model's endemic-state initialisation discards.

Outbreak-responsive management. In clinical practice, a newly detected AMR cluster triggers immediate responses: enhanced surveillance, patient cohorting, temporary transfer restrictions, and emergency hand hygiene campaigns. The inter-ward transfer rates T_{ij} — set from long-run statistics — are typically zeroed during an active outbreak as clinical teams impose transfer moratoriums. Our static-parameter model cannot represent this adaptive response, and would therefore systematically overestimate spread to other wards.

Practical implication: The model should be explicitly labelled as a **steady-state policy planning tool** for endemic AMR management. It answers the question: "Given entrenched AMR colonisation, which sustained intervention programme reduces endemic prevalence most?" It does not answer: "What will happen if one colonised patient arrives tomorrow?" — a question requiring stochastic outbreak modelling with time-varying, response-adaptive parameters.

8 Conclusion

This study developed and analysed a four-ward coupled compartmental model of antimicrobial resistance transmission in a hospital network, combining a Susceptible–Colonised–Susceptible patient framework with explicit healthcare worker contamination dynamics.

Key findings:

1. **Endemic persistence is driven by admission colonisation forcing:** The system-level basic reproduction number $R_0 = 0.96 < 1$, meaning intrinsic in-hospital transmission alone would not sustain AMR. Endemic persistence arises instead from the constant admission colonisation forcing term ($\alpha\mu N$), which continuously seeds colonised patients regardless of in-hospital transmission intensity. Reducing R_0 further through hand hygiene or antibiotic stewardship cannot eliminate endemic AMR without simultaneously controlling admission colonisation.

2. **Transmission and imported colonisation jointly dominate endemic prevalence:** Sensitivity analysis reveals that the transmission rate β (PRCC 0.95) and

admission colonisation rate α (PRCC 0.92) are the two strongest drivers of 12-month prevalence, while α remains negligible in R_0. This asymmetry means that R_0 alone is an incomplete guide to intervention priority.

3. **Scenario B (admission screening) achieves the largest prevalence reduction:** Reducing admission colonisation by 80%, combined with 30% reduction in direct transmission, reduces system-wide prevalence from 7.3% to 1.0%. This dramatically outperforms interventions targeting only in-hospital transmission (Scenarios A and C).

4. **Combined Scenario D achieves the lowest R_0:** The combined hand hygiene + antibiotic stewardship strategy ($R_0 = 0.88$) best reduces the risk of epidemic amplification during outbreak events, even though its effect on endemic prevalence (6.3%) is far inferior to Scenario B (1.0%) because it does not address the admission colonisation forcing term.

Policy recommendation to the hospital board: Implement universal admission screening and contact precautions (Scenario B) as the primary intervention for reducing endemic AMR burden. Antibiotic stewardship (Scenario C) remains a separately supported follow-on option for sustained management of antibiotic selection pressure, but the present model did not simulate a dedicated B+C package. Hand hygiene improvement (Scenario A) should be maintained as a baseline standard given its structural effect on R_0 and its low implementation cost. The combined A+C strategy (Scenario D) is recommended for periods of elevated outbreak risk.

Limitations: The most significant are the homogeneous mixing assumption, deterministic formulation for small wards (especially ICU with $N = 15$), and time-invariant parameters. These are discussed fully in Section 7. The model is appropriate for steady-state endemic planning and should not be applied to acute outbreak scenarios.

Future work: Priority refinements include stochastic simulation for the ICU ward, time-varying seasonal parameters, and extension to a two-strain model to evaluate antibiotic cycling strategies on a multi-year horizon.

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